

Problem Solving Using Python (CSE1012)

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Dictionary

- Dictionary is a data structure in which we store values as pair of key and value
- Each key is separated from its value by a colon (:)
- And consecutive items are separated by commas
- Entire items are enclosed in {}

Dictionary

- Keys must be unique and be of any immutable data types (like strings, numbers, or tuples)
- There is not stringent requirement for uniqueness and type of values
- i.e. value of a key can be of any type

Dictionary

- Dictionaries are not sequences
- They are mappings
- Mappings are collections of objects that store objects by key instead of by relative position
- Keys are case-sensitive

Creating a Dictionary

```
dictionary_name = {k1: val1, k2: val2, k3: val3}
```

```
Dict = {}          #Creates empty dictionary
```

```
Dict = {'Roll': '01', 'Name': 'Arnab', 'Course': 'Btech'}
```

Creating a Dictionary

#dictionary with keys as number

```
Dict = {x: 2*x for x in range(1,11)}
```

```
print(Dict)
```

```
{1: 2, 2: 4, 3: 6, 4: 8, 5: 10, 6: 12...}
```

Creating a Dictionary

```
Dict = {'Roll_No' : '16/001', 'Name' : 'Arav', 'Course' : 'BTech'}  
print(Dict)
```

OUTPUT

```
{'Roll_No': '16/001', 'Name': 'Arav', 'Course': 'BTech'}
```


Accessing values in a Dictionary

- To access values in a dictionary, square brackets are used along with the key to obtain its value

Accessing values in a Dictionary

```
Dict = {1: 2, 2: 4, 3: 6}
```

```
print(Dict[2]) # here 2 is a key in the dictionary.
```

```
4             # value saved under the key is 4.
```

Accessing values in a Dictionary

```
Dict = {'Roll_No' : '16/001', 'Name' : 'Arav', 'Course' : 'BTech'}  
print("Dict[ROLL_NO] = ", Dict['Roll_No'])  
print("Dict[NAME] = ", Dict['Name'])  
print("Dict[COURSE] = ", Dict['Course'])
```

OUTPUT

```
Dict[ROLL_NO] = 16/001  
Dict[NAME] = Arav  
Dict[COURSE] = BTech
```

Adding an entry to the Dictionary

- To add a new entry or a key-value pair in a dictionary, just specify the key-value pair

```
Dict = {'Roll_No': '16/001', 'Name': 'Arnav', 'Course': 'Btech'}
```

```
Dict['Marks']= 95
```

```
print(Dict)
```

Modifying an entry in the Dictionary

- To modify an entry, just overwrite the existing value

```
Dict = {'Roll_No': '16/001', 'Name': 'Arnav', 'Course': 'Btech'}
```

```
Dict['Marks']= 95      #Adding new entry
```

```
Dict['Course']='BCA'   #Modifying an entry
```

Modifying an entry in the Dictionary

```
Dict = {'Roll_No' : '16/001', 'Name' : 'Arav', 'Course' : 'BTech'}  
print("Dict[ROLL_NO] = ", Dict['Roll_No'])  
print("Dict[NAME] = ", Dict['Name'])  
print("Dict[COURSE] = ", Dict['Course'])  
Dict['Marks'] = 95      # new entry  
print("Dict[MARKS] = ", Dict['Marks'])
```

OUTPUT

```
Dict[ROLL_NO] =  16/001  
Dict[NAME] =  Arav  
Dict[COURSE] =  BTech  
Dict[MARKS] =  95
```

Programming Tip: Trying to index a key that isn't part of the dictionary returns a `KeyError`.

Modifying an entry in the Dictionary

```
Dict = {'Roll_No' : '16/001', 'Name' : 'Arav', 'Course' : 'BTech'}  
print("Dict[ROLL_NO] = ", Dict['Roll_No'])  
print("Dict[NAME] = ", Dict['Name'])  
print("Dict[COURSE] = ", Dict['Course'])  
Dict['Marks'] = 95      # new entry  
print("Dict[MARKS] = ", Dict['Marks'])  
  
Dict['Course'] = 'BCA'  
print("Dict[COURSE] = ", Dict['Course']) #entry updated
```

OUTPUT

```
Dict[ROLL_NO] = 16/001  
Dict[NAME] = Arav  
Dict[COURSE] = BTech  
Dict[MARKS] = 95  
Dict[COURSE] = BCA
```

Deleting an entry in the Dictionary

- You can delete an item from the dictionary using del keyword
- To delete all items in just one statement, use the clear function as: `dict_name.clear()`
- Finally, to delete the entire dictionary from the memory, use: `del dict_name`

Deleting an entry in the Dictionary

```
dict={'Roll': '16/001', 'Name': 'Arnav', 'Course': 'Btech'}  
print(dict)
```

```
del dict['Course']      #deletes a key value pair  
print("After deleting Course: ",dict)
```

Deleting an entry in the Dictionary

```
dict={'Roll': '16/001', 'Name': 'Arnav', 'Course': 'Btech'}  
print(dict)
```

```
dict.clear()                #deletes all entries in the dictionary  
print("After clear(), the dictionary has no item: ",dict)
```

```
del dict                    # deletes the variable dict from memory
```

Deleting an entry in the Dictionary

- You can also use the `pop()` method to delete a particular key from the dictionary

`dict.pop(key)`

- The `pop` method removes an item from the dictionary and returns its value

Deleting an entry in the Dictionary

```
dict.pop(key[,default])
```

- The pop method removes an item from the dictionary and returns its value
- If the specified key is not present, then the default value is returned
- The default is optional. But if you don't specify it and the key is also not there, there will be error

Deleting an entry in the Dictionary

```
dict = {'Roll': '16/001', 'Name': 'Arnav', 'Course': 'Btech'}  
dict.pop('Name')  
'Arnav'
```

Deleting an entry in the Dictionary

```
dict={'Roll': '16/001', 'Name': 'Arnav', 'Course': 'Btech'}
```

```
dict.pop('Marks',-1)
```

```
-1
```

Deleting an entry in the Dictionary

```
dict={'Roll': '16/001', 'Name': 'Arnav', 'Course': 'Btech'}  
dict.pop('Marks')      #shows following error,
```

Traceback (most recent call last):

File "<pyshell#13>", line 1, in <module>

dic.pop('Marks')

KeyError: 'Marks'

Key points to Remember

- Keys must have unique values
- If you try to add a duplicate key, then the last assignment is retained

Key points to Remember

- Keys must be of a type that is immutable
- i.e. a key can be of strings, number, or tuple
- But a key can't be a list which is mutable

Key points to Remember

```
dic = {(1, [1, 2, 3]): 100}    # trying to make tuple as a key with  
                                # a list within it  
                                # it shows following error
```

Traceback (most recent call last):

```
File "<pyshell#59>", line 1, in <module>
```

```
dic={(1,[1,2,3]): 100}
```

```
TypeError: unhashable type: 'list'
```

Searching a key in the Dictionary

```
dict = {'Roll': '16/001', 'Name': 'Arnav', 'Course': 'Btech'}
```

```
if 'Course' in dict:
```

```
    print(dict['Course'])
```

Output:

Btech

Sorting items in a Dictionary

```
dict = {'Roll': '16/001', 'Name': 'Arnav', 'Course': 'Btech'}  
print(dict.keys())  
dict_keys(['Roll', 'Name', 'Course'])
```

```
print(sorted(dict.keys()))  
['Course', 'Name', 'Roll']
```

Looping over a Dictionary

- You can loop over a dictionary to access the values, keys, and both keys and values

Looping over a Dictionary

```
dict = {'Roll': '16/001', 'Name': 'Arnav', 'Course': 'Btech'}  
print("Keys: ",end=" ")  
for key in dict:  
    print(key, end=" ")
```

Output:

Roll Name Course

Looping over a Dictionary

```
dict= {'Roll': '16/001', 'Name': 'Arnav', 'Course': 'Btech'}  
print("Values: ", end=" ")  
for val in dict.values():  
    print(val, end=" ")
```

Output:

16/001 Arnav Btech

Looping over a Dictionary

```
dict= {'Roll': '16/001', 'Name': 'Arnav', 'Course': 'Btech'}  
print("Dictionary: ", end=" ")  
for key, val in dict.items():  
    print(key, val, "\t", end=" ")
```

Output:

Roll 16/001	Name Arnav	Course Btech
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Nested Dictionary

- You can define a dictionary within another dictionary

Nested Dictionary

```
#prog to demonstrate nested dictionary
students={'Shiv': {'Py': 90, 'DSA': 89, 'DBMS': 92},
          'Sadhvi': {'Py': 91, 'DSA': 87, 'DBMS': 94},
          'Krish': {'Py': 93, 'DSA': 92, 'DBMS': 88}},
for key, val in students.items():
    print(key, val)
```

Dictionary functions & Methods

Operation	Description	Example	Output
<code>len(Dict)</code>	Returns the length of dictionary. That is, number of items (key-value pairs)	<pre>Dict1 = {'Roll_No' : '16/001', 'Name' : 'Arav', 'Course' : 'BTech'} print(len(Dict1))</pre>	3
<code>str(Dict)</code>	Returns a string representation of the dictionary	<pre>Dict1 = {'Roll_No' : '16/001', 'Name' : 'Arav', 'Course' : 'BTech'} print(str(Dict1))</pre>	<pre>{'Name': 'Arav', 'Roll_No': '16/001', 'Course': 'BTech'}</pre>
<code>Dict.clear()</code>	Deletes all entries in the dictionary	<pre>Dict1 = {'Roll_No' : '16/001', 'Name' : 'Arav', 'Course' : 'BTech'} Dict1.clear() print(Dict1)</pre>	<pre>{}</pre>
<code>Dict.copy()</code>	Returns a shallow copy of the dictionary, i.e., the dictionary returned will not have a duplicate copy of Dict but will have the same reference	<pre>Dict1 = {'Roll_No' : '16/001', 'Name' : 'Arav', 'Course' : 'BTech'} Dict2 = Dict1.copy() print("Dict2 : ", Dict2) Dict2['Name'] = 'Saesha' print("Dict1 after modification : ", Dict1)</pre>	<pre>Dict2 : {'Course': 'BTech', 'Name': 'Arav', 'Roll_No': '16/001'} Dict1 after modification: {'Course': 'BTech',</pre>

Dictionary functions & Methods

		<pre>print("Dict2 after modification :",Dict2)</pre>	<pre>'Name': 'Arav', 'Roll_No': '16/001'} Dict2 after modification: {'Course': 'BTech', 'Name': 'Saasha', 'Roll_No': '16/001'}</pre>
<code>Dict.fromkeys(seq[,val])</code>	Create a new dictionary with keys from seq and values set to val. If no val is specified then, None is assigned as default value	<pre>Subjects = ['CSA', 'C++', 'DS', 'OS'] Marks = dict. fromkeys(Subjects, -1) print(Marks)</pre>	<pre>{'OS': -1, 'DS': -1, 'CSA': -1, 'C++': -1}</pre>
<code>Dict.get(key)</code>	Returns the value for the key passed as argument. If the key is not present in dictionary, it will return the default value. If no default value is specified then it will return None	<pre>Dict1 = {'Roll_No' : '16/001', 'Name' : 'Arav', 'Course' : 'BTech'} print(Dict1.get('Name'))</pre>	Arav
<code>Dict.has_key(key)</code>	Returns True if the key is present in the dictionary and False otherwise	<pre>Dict1 = {'Roll_No' : '16/001', 'Name' : 'Arav', 'Course' : 'BTech'} print('Marks' in Dict1)</pre>	False
<code>Dict.items()</code>	Returns a list of tuples (key-value pair)	<pre>Dict1 = {'Roll_No' : '16/001', 'Name' : 'Arav', 'Course' : 'BTech'} print(Dict1.items())</pre>	<pre>[('Course', 'BTech'), ('Name', 'Arav'), ('Roll_No', '16/001')]</pre>
<code>Dict.keys()</code>	Returns a list of keys in the dictionary	<pre>Dict1 = {'Roll_No' : '16/001', 'Name' : 'Arav', 'Course' : 'BTech'} print(Dict1.keys())</pre>	<pre>['Course', 'Name', 'Roll_No']</pre>
<code>Dict.setdefault(key, value)</code>	Sets a default value for a key that is not present in the dictionary	<pre>Dict1 = {'Roll_No' : '16/001', 'Name' : 'Arav', 'Course' : 'BTech'}</pre>	Arav has got marks = 0

Dictionary functions & Methods

		<pre>Dict1. setdefault('Marks',0) print(Dict1['Name'], "has got marks = ", Dict1. get('Marks'))</pre>	
<code>Dict1.update(Dict2)</code>	Adds the key-value pairs of Dict2 to the key-value pairs of Dict1	<pre>Dict1 = {'Roll_No' : '16/001', 'Name' : 'Arav', 'Course' : 'BTech'} Dict2 = {'Marks' : 90, 'Grade' : 'O'} Dict1.update(Dict2) print(Dict1)</pre>	<pre>{'Grade': 'O', 'Course': 'BTech', 'Name': 'Arav', 'Roll_No': '16/001', 'Marks': 90}</pre>
<code>Dict.values()</code>	Returns a list of values in dictionary	<pre>Dict1 = {'Roll_No' : '16/001', 'Name' : 'Arav', 'Course' : 'BTech'} print(Dict1.values())</pre>	<pre>['BTech', 'Arav', '16/001']</pre>
<code>Dict.items()</code>	Used to iterate through items in the dictionary	<pre>Dict = {'Roll_No' : '16/001', 'Name' : 'Arav', 'Course' : 'BTech'} for i,j in Dict. items(): print(i, j)</pre>	<pre>Course BTech Name Arav Roll_No 16/001</pre>
<code>in and not in</code>	Checks whether a given key is present in dictionary or not	<pre>Dict = {'Roll_No' : '16/001', 'Name' : 'Arav', 'Course' : 'BTech'} print('Name' in Dict) print('Marks' in Dict)</pre>	<pre>True False</pre>

Difference between Dictionary & List

- There are some important differences
- **First**, a list is an ordered items. But, a dictionary is a data structure that is used for matching one item (key) with another (value)

Difference between Dictionary & List

- **Second**, in lists, you can use indexing to access items. These indexes should be numbers
- In dictionaries, you can use any immutable type of object as an index
- Ex, dict['Name']

Difference between Dictionary & List

- **Third**, lists are used to look up a value whereas a dictionary is used to take one value and look up another value
- Dictionaries are also called a lookup table

Difference between Dictionary & List

- **Fourth**, the key-value pair may not be displayed in the order in which it was specified while defining the dictionary
- This is because, Python uses complex algorithm (called hashing) to provide fast access to the items stored in the dictionary

String formatting with Dictionaries

- Python also allows you to use string formatting feature with dictionaries
- So, you can use %s, %d, %f, etc. to represent string, integer, floating point number, or any other data

String formatting with Dictionaries

- Example: Program that uses string formatting feature to print the key-value pairs stored in the dictionary

```
Dict = {'Sneha' : 'BTech', 'Mayank' : 'BCA'}  
for key, val in Dict.items():  
    print("%s is studying %s" % (key, val))
```

OUTPUT

```
Sneha is studying BTech  
Mayank is studying BCA
```

When to use which Data Structure?

- Use **List** to store a collection of data that doesn't need random access
- Use **List** if the data has to be modified frequently

When to use which Data Structure?

- Use a **set** if you want to ensure that every element in the data structure must be unique
- Use **tuples** when you want that your data shouldn't be altered

When to use which Data Structure?

- **Dictionary** is used to store key-value pairs
- If you need frequent lookup operations, use **dictionaries**

Thank You

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